

GLEB VORONKOV

AI / ML Engineer for Mobile Core Networks (4G · 5G) · Exploring AI-native 6G

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PROFESSIONAL SUMMARY

Telecom AI/ML engineer focused on predictive analytics, automation, and decision-support for **4G EPC / 5G Core (PS Core)**. Experience with Huawei USN/UGW, GTP-U / PFCP / Diameter, KPI telemetry, and SON-style forecasting (paging, traffic, load). Building LLM assistants that generate network-element configurations from 3GPP and vendor documentation. Expanding toward **AI-native network intelligence** and **6G research themes** relevant to Ericsson, Huawei, Cisco, and ZTE.

CORE COMPETENCIES

AI for Mobile	Traffic / paging / load forecasting (LSTM, Temporal Fusion Transformer, Prophet, XGBoost); VAE anomaly detection; Contextual Bandits & RL for policy control; SHAP; AutoML; LLM/RAG for ops.
PS Core / RAN	4G EPC (MME, SGW, PGW, Policy Control, HSS, USN/UGW); 5G Core (AMF, SMF, UPF); NR/LTE traffic steering; GTP-U, PFCP, Diameter; YANG; policy-rules; session tracing & scaling. Radio: QAM/OFDMA/QPSK, LTE/5G; studying AI-assisted 6G concepts.
ML / MLOps	TensorFlow, PyTorch, scikit-learn, Keras, MLflow, Kubeflow; Docker, Kubernetes, Helm; FastAPI/Flask; GitLab CI / Jenkins; Airflow; Prometheus, Grafana, Elastic Stack.
Data / Code	pandas, NumPy, Dask, Spark, Kafka; PostgreSQL, MySQL, MongoDB, SQLite; Python 3.10+, C++, C#, JavaScript/Node.js, Bash, SQL, YANG.

PROFESSIONAL EXPERIENCE

AI / ML Systems & Solutions Engineer

Telcore · Jan 2026 – Present · Moscow

- Design AI/ML architectures for telecom use cases; train/validate models; monitor deployed solution reliability and scale.
- Lead **AI PS Core Assistant**: RAG over 3GPP/internal docs, Q&A, JSON/YAML/CLI config generation, UML procedure diagrams.
- Build AI-assisted QA test generators to accelerate automated testing for network software and tooling.

PS Core Engineer (AI & Automation)

VimpelCom (Beeline) · Jun 2025 – Present · Moscow

- Prototype ML models for traffic, paging, and core-network load forecasting to improve congestion readiness.
- Automate optimization/ops workflows (Python, Node.js, Bash); build datasets from KPIs (CPU, GTP-U, NAS, UE attach).
- Develop/test PS Core configs (policy-rules, PFCP, scaling) on **Huawei USN/UGW**; document AI adoption across EPC/5GC.

Project Lead — ARTCam Virtual Content Studio

Research Division, MTUCI (Moscow Technical University of Communications and Informatics) · Jan 2023 – Present

- Lead R&D for motion-capture / 3D pipelines; registered software for video-to-3D motion capture in Unreal Engine (No. 2024682445).
- Publish peer-reviewed research (RSCI / Scopus) on XR content production systems.

Independent Projects — AI for Telecom & XR

Aug 2021 – Feb 2023

- AI forecasting for eNB load and energy consumption; Unreal Engine 4/5 C++ tooling for VR/AR; web prototypes (JS/Python).

SELECTED TELECOM AI PROJECTS

SON Paging Prediction (Beeline)

Lead — LSTM + Spark + Docker/K8s + PostgreSQL. 7-day horizon, MAPE < 5%, proactive capacity actions before paging overload.

NR/LTE Traffic Separation (Beeline / Huawei USN-UGW)

Lead — CLI automation on Huawei PS Core. NR/LTE traffic marking and A/B validation in a test network.

AI Engineer Assistant for PS Core (Beeline)

Lead — Python, TensorFlow, Qwen2.5-7B, Docker. LLM assistant proposes PS Core configuration variants from large doc corpora.

AI Task Matrix for EPC + AI PS Core Assistant (Beeline / Telcore)

Mapped prioritized AI use cases across MME/SGW/PGW/PCEF/HSS/OSS; RAG assistant + AI test-code generation for telecom QA.

EDUCATION

M.Sc. Radio Engineering — Media Content Processing Systems · MTUCI · 2025–2027

AI Systems; Mobile Network Optimization; Probability & Stochastic Processes; Advanced Math & Linear Algebra.

B.Sc. Infocommunication Technologies and Communication Systems · MTUCI · 2021–2025

LTE/5G Networks; Telecommunication Protocols; ML Algorithms for Telecom; Graph Theory & Coding.

PUBLICATIONS & OPEN SOURCE

- Voronkov G.A., Vinetsky V.Yu. Character Motion Capture for Broadcast Content Production. Sync./Signal Processing Systems, 2025 (RSCI).
- Voronkov G.A., Matskevich A.G., Orlov V.G. Virtual Video Content Studio Model with AR. Telecom & IT, 2023 (RSCI).
- V. G. Voronkov — Scopus papers on XR content assessment (2024) and 3D model generation error analysis (2025).
- GitHub: github.com/GlebVoronkov03 — MLOps pipelines, VAE detectors, TFT forecasting models.

Target roles: AI Engineer / PS Core / Cloud-Native Network Intelligence at global vendors & operators (Ericsson, Huawei, Cisco, ZTE) — 4G/5G core AI, SON/automation, AI-readiness for 6G.